

Sai Maheswar Babbellapati Venkata

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Summary

Full-stack and AI engineer with 5+ years of experience building Python web applications, backend systems, and machine learning services. Skilled in Python, FastAPI, Flask, NumPy, and agentic engineering workflows, with hands-on experience deploying containerized applications across AWS, Azure, and GCP using Docker and Kubernetes. Proven background optimizing SQL and NoSQL databases, building REST APIs, and collaborating with cross-functional teams to ship reliable software.

Experience

Python Full Stack Engineer

USA

Truist Bank

July 2024 – present

- Integrated AI-powered APIs and machine learning services into banking workflows to automate document validation and customer data processing, saving operations teams 10+ hours weekly that used to go into manual review.
- Owned end-to-end delivery of backend services using Python, FastAPI, Flask, Go, and REST APIs, breaking large banking applications into scalable microservices that processed 50K+ API requests daily for enterprise stakeholders.
- Tuned PostgreSQL queries by redesigning indexes, optimizing joins, and caching frequently accessed data in Redis, reducing peak query latency by 12% while supporting 10K+ customer records during month-end processing spikes.
- Architected CI/CD pipelines using Docker, Kubernetes, Jenkins, GitHub Actions, and GitLab CI/CD, enabling automated production deployments that cut release windows from 2 hours to under 20 minutes.
- Improved production reliability by implementing centralized logging, health checks, retry mechanisms, and exception handling, reducing recurring production incidents by nearly 20%.
- Built responsive web applications with Angular, TypeScript, JavaScript, HTML5, and CSS3, creating reusable components and integrating secure REST APIs for desktop and mobile views.
- Partnered directly with solution architects, security engineers, QA teams, and business stakeholders to translate requirements into secure applications supporting multi-million-dollar transaction processing.

Junior Python Full Stack Engineer

India

NVIDIA

June 2020 – Apr 2023

- Designed and built backend services using Python, FastAPI, Flask, C++, and Go, developing REST APIs and scalable architectures supporting engineering platforms used by 50+ internal developers.
- Used CUDA, TensorRT, and NVIDIA GPU frameworks to integrate GPU-accelerated AI workflows into internal applications, reducing processing time for compute-intensive workloads by ~20%.
- Integrated machine learning models with backend inference services and data pipelines, simplifying production deployment and cutting engineering integration effort by nearly 30%.
- Deployed containerized, cloud-native applications using Docker, Kubernetes, and Helm, maintaining consistent production releases across development, QA, and enterprise environments.
- Optimized distributed data systems including PostgreSQL, MongoDB, Redis, and Elasticsearch, introducing indexing strategies that reduced search response times while managing millions of records.
- Automated build and deployment pipelines with Jenkins, GitLab CI/CD, Git, and Linux scripting, eliminating repetitive manual tasks and reducing release preparation time by 25%.
- Developed modern web interfaces with React.js, TypeScript, JavaScript, HTML5, and CSS3, creating 10+ reusable UI components across engineering dashboards.

Projects

Cloud-Native AI-Powered E-Commerce Platform (Microservices Architecture)

- Technologies: Python, FastAPI, React.js, PostgreSQL, MongoDB, Docker, Kubernetes, AWS
- Built a full-stack, cloud-native application from scratch with a Python FastAPI backend and a React.js frontend, implementing REST APIs for catalog browsing, order processing, and customer account management.
- Designed a microservices architecture with Docker containerization and Kubernetes deployment on AWS, integrating PostgreSQL for transactional data and MongoDB for product catalog documents with custom indexing for fast search.

AI-Driven Predictive Analytics Platform

- Technologies: Python, Scikit-learn, Pandas, Flask, React.js, SQL, Docker
- Built a predictive analytics application that takes in raw tabular data and outputs predictions using Python, Pandas, NumPy, and scikit-learn for data preprocessing, feature engineering, and model training.
- Developed REST APIs with Flask to serve predictions to a React frontend and containerized the system with Docker to maintain execution consistency across environments.

Education

Southern Arkansas University
MS in Computer Information Science

Magnolia, AR
May 2023 – Dec 2024

GITAM
BS in Computer Science

Hyderabad, India
May 2018 – May 2022

Skills

Programming: Python, C++, Go, Java, JavaScript, TypeScript, SQL, Bash

AI/ML & Agentic Tools: Machine Learning, NumPy, Scikit-learn, PyTorch, TensorFlow, NLP Frameworks, Transformers (GPT, BERT), LangChain, Agentic Engineering Workflows, Model Serving, NVIDIA GPU Computing, CUDA, TensorRT

Backend & APIs: FastAPI, Flask, Django, REST APIs, GraphQL, Microservices, API Gateway, Async Programming

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Oracle, SQL Server, DynamoDB, NoSQL

Cloud & DevOps: AWS (EC2, S3, Lambda, ECS, EKS), Azure (AKS, Functions, Blob Storage), GCP, Docker, Kubernetes, Helm, Jenkins, GitHub Actions, GitLab CI/CD, Terraform

Frontend: React.js, Angular, TypeScript, JavaScript, HTML5, CSS3, Redux, Material UI

Systems & Testing: Distributed Systems, System Design, Data Structures & Algorithms, PyTest, Unit/Integration Testing, Postman, ELK Stack, Grafana, OAuth2, JWT